

A BIVARIATE EXTENSION OF THE GAMMA DISTRIBUTION

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Key words: Bivariate extension, exponential, gamma, Poisson process.

SUMMARY

A bivariate extension of the gamma distribution is derived. It includes as a special case the well-known bivariate extension of the exponential distribution derived by Freund (1961). The moments of and parameter estimation for the bivariate gamma extension is also dealt with.

1. Introduction

A number of bivariate exponential distributions and bivariate exponential extensions have been introduced by various authors such as Gumbel (1960), Freund (1961), Downton (1970), Marshall and Olkin (1967), Block and Basu (1974), Block (1975) and Friday and Patil (1977). Bivariate exponential distributions are bivariate distributions with exponential marginal distributions which are generally not independent and were not used in the derivation either. Bivariate exponential extensions on the other hand do not have exponential marginal distributions but are derived from independent exponential distributions.

A number of bivariate gamma distributions have been reported in the literature. Most of these distributions are difficult to interpret. Furthermore none of these bivariate gamma distributions contain any of the mentioned bivariate exponential distributions or extensions as a special case.

In this paper the following approach is used to find a density function for a sensible bivariate gamma extension. Interpret the bivariate exponential extension (Freund, 1961) in terms of a shock model i.e. each component fails after receiving a single shock whose arrival is governed by an independent Poisson process when both components still function. However, when one component fails the parameter of the Poisson process governing the arrival of the shock to the still functioning component changes. A bivariate gamma extension can then be derived making use of the fact that the waiting time for a number of shocks whose arrivals are governed by a Poisson process has a gamma distribution.

2. The Bivariate Gamma Extension

The density function of the bivariate gamma extension derives from the following

model. Let the random variables X and Y denote the lifetimes of the components C_1 and C_2 in a two component system where C_1 fails after receiving h shocks and C_2 fails after receiving l shocks and while both components are still functioning these shocks are governed by two independent Poisson processes with parameters $1/\alpha$ and $1/\beta$ respectively. As long as both components are still functioning their lifetimes are independent, but the moment one fails the changed load on the remaining component results in a change of the parameter of the Poisson process governing the shocks to the still functioning component i.e. $1/\alpha(1/\beta)$ changes to $1/\alpha'(1/\beta')$ when $C_2(C_1)$ fails first.

It is convenient to find the density function of the bivariate gamma extensions from its survival function i.e.

$$(2.1) \quad f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} \bar{F}_{X,Y}(x,y)$$

where $\bar{F}_{X,Y}(x,y) = P(X > x, Y > y)$ and it may hold for x and y that either $0 < x < y$ or $0 < y < x$.

For $0 < x < y$ the survival function may now be written as

$$\begin{aligned} \bar{F}_{X,Y}(x,y) = & P(x < X < y, Y > y; \quad C_1 \text{ fails before } C_2) \\ & + P(X > y, Y > y; \quad C_1 \text{ fails before } C_2) \\ & + P(X > y, Y > y; \quad C_2 \text{ fails before } C_1) \end{aligned}$$

so that (2.1) becomes

$$(2.2) \quad f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} P(x < X < y, Y > y; C_1 \text{ fails before } C_2)$$

since the other probabilities are functions of y only.

Now

$$(2.3) \quad P(x < X < y, Y > y; C_1 \text{ fails before } C_2)$$

$$= \int_x^y P(Y > y | X=u) f_X(u) du$$

$$= \int_x^y \sum_{j=0}^{l-1} \sum_{i=0}^{l-1-j} \frac{\left(\frac{u}{\beta}\right)^j}{j!} e^{-\frac{u}{\beta}} \frac{\left(\frac{y-u}{\beta'}\right)^i}{i!} e^{-\frac{(y-u)}{\beta'}} \frac{u^{h-1} e^{-\frac{u}{\alpha}}}{\alpha^h \Gamma(h)} du$$

since the waiting time of C_1 to receive h shocks has a gamma distribution with parameters h and α , and

$$\begin{aligned} P(Y > y | X = u) &= P(C_2 \text{ functions longer than time } y | C_1 \text{ failed at time } u) \\ &= P(C_2 \text{ receives at most } \ell-1 \text{ shocks before time } y \text{ of which } j \text{ were before time } u \text{ and were governed by a Poisson process with parameter } 1/\beta \text{ and } i \text{ of them were between time } u \text{ and time } y \text{ and were governed by a Poisson process with parameter } 1/\beta' \text{ and } i+j \leq \ell-1). \end{aligned}$$

Employing $\Gamma(m, x) = \int_0^x u^{m-1} e^{-u} du$ and

$$(2.4) \quad (y-u)^i = \sum_{k=0}^i \binom{i}{k} y^k (-u)^{i-k}$$

it follows from (2.3) that

$$(2.5) \quad P(x < X < y, Y > y; C_1 \text{ fails before } C_2)$$

$$\begin{aligned} &= \frac{\alpha^{-h} e^{-y}}{\Gamma(h)} \sum_{j=0}^{\ell-1} \sum_{i=0}^{\ell-1-j} \sum_{k=0}^i \frac{\beta^{-j} (\beta')^{-i} (-1)^{i-k} y^k}{j! k! (i-k)!} \left(\frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{\beta'} \right)^{-(h+j+i-k)} \\ &\quad \cdot \{ \Gamma[h+j+i-k; (\frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{\beta'})y] - \Gamma[h+j+i-k; (\frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{\beta'})x] \} \end{aligned}$$

Similarly for $0 < y < x$ it follows that

$$(2.6) \quad P(X > x, y < Y < x; C_2 \text{ fails before } C_1)$$

$$\begin{aligned} &= \frac{\beta^{-\ell}}{\Gamma(\ell)} e^{-x} \sum_{j=0}^{h-1} \sum_{i=0}^{h-1-j} \sum_{k=0}^i \frac{\alpha^{-j} (\alpha')^{-i} (-1)^{i-k} x^k}{j! k! (i-k)!} \left(\frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{\alpha'} \right)^{-(\ell+j+i-k)} \\ &\quad \cdot \{ \Gamma[\ell+j+i-k; (\frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{\alpha'})x] - \Gamma[\ell+j+i-k; (\frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{\alpha'})y] \} \end{aligned}$$

After some algebraic manipulations the joint density function of X and Y follows from (2.2), (2.5) and (2.6) as

$$(2.7) \quad f_{X,Y}(x,y) = \begin{cases} \frac{\alpha^{-h}(\beta')^{-1}}{\Gamma(h)\Gamma(\ell)} x^{h-1} \left[\frac{1}{\beta'}(y-x) + \frac{1}{\beta}x \right]^{\ell-1} \\ \cdot e^{-\frac{y}{\beta'}} e^{-\left(\frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{\beta'}\right)x} & \text{for } 0 < x < y \\ \frac{\beta^{-\ell}(\alpha')^{-1}}{\Gamma(h)\Gamma(\ell)} y^{\ell-1} \left[\frac{1}{\alpha'}(x-y) + \frac{1}{\alpha}y \right]^{h-1} \\ \cdot e^{-\frac{x}{\alpha'}} e^{-\left(\frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{\alpha'}\right)y} & \text{for } 0 < y < x. \end{cases}$$

Remarks:

1. For $h = \ell = 1$ the bivariate exponential extension of Freund is a special case of the bivariate gamma extension.
2. Figure 2.1 gives a graphical representation of the bivariate gamma extension in the special case $h = \ell = 2$, $\alpha = \beta = 1$ and $\alpha' = \beta' = 4$.
3. The marginal distribution of X in the bivariate gamma extension is

$$\begin{aligned} f_X(x) &= \frac{\alpha^{-h}}{\Gamma(h)} \sum_{k_1=0}^{\ell-1} \frac{\beta^{-k_1}}{k_1!} x^{h+k_1-1} e^{-\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)x} \\ &+ \frac{\frac{\alpha\alpha'}{\beta(\alpha'-\alpha)}}{\Gamma(\ell)\alpha'} \sum_{k_2=0}^{h-1} \frac{\Gamma[h+\ell-k_2-1; \left(\frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{\alpha'}\right)x]}{(h-1-k_2)! k_2!} \\ &\cdot \left[1 + \frac{\alpha\alpha'}{\beta(\alpha'-\alpha)} \right]^{-\left(h+\ell-k_2-1\right)} \left(\frac{x}{\alpha'} \right)^{k_2} e^{-\frac{x}{\alpha'}} \\ &\quad \text{for } \frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{\alpha'} > 0 \end{aligned}$$

and by symmetry the marginal distribution of Y follows from $f_X(x)$. First and second order moments for these distributions follow in section 3.

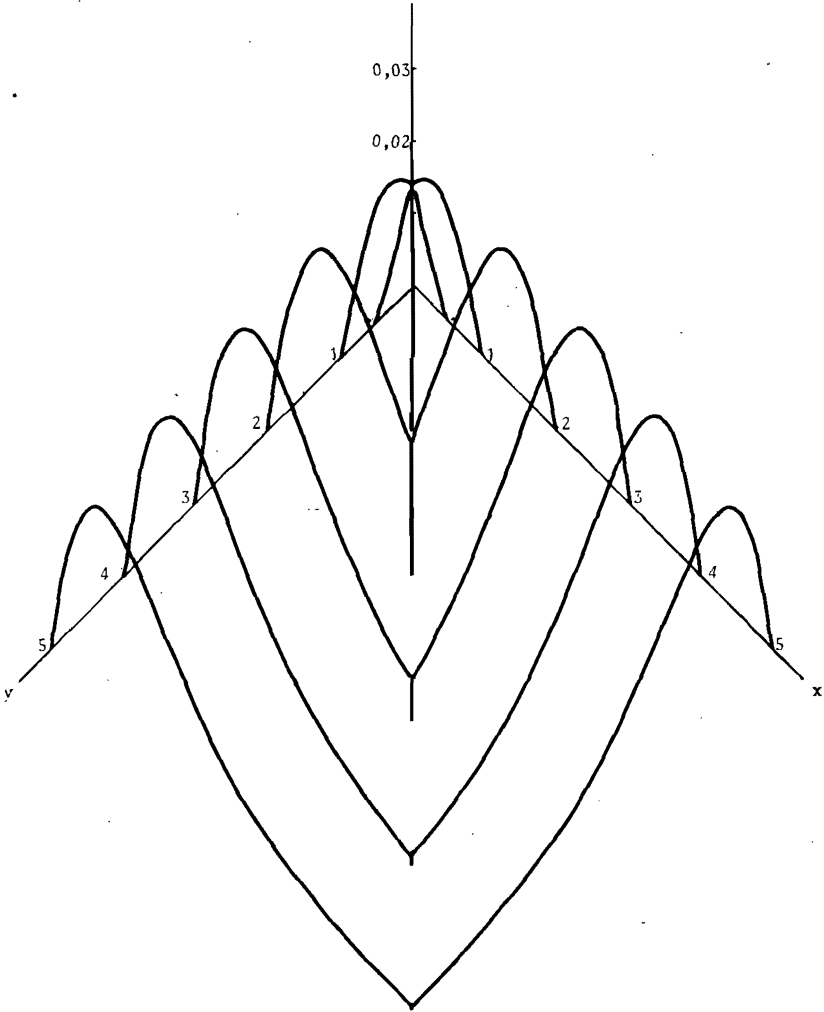


Figure 2.1

3. The Moment Generating Function

The moment generating function of (2.7) is defined by

(3.1) $M_{X,Y}(s,t) = E^I [e^{sX + tY}] + E^{II} [e^{sX + tY}]$

where I and II respectively denote the regions $0 < x < y$ and $0 < y < x$.

From (2.7) for $0 < x < y$

$$(3.2) \quad E^I[e^{sX+tY}] = \frac{\alpha^{-h}(\beta')^{-\ell}}{\Gamma(h)} \sum_{k_1=0}^{\ell-1} \frac{(\frac{\beta'}{\beta})^{k_1}}{k_1!} \frac{1}{\Gamma(\ell-k_1)} \\ \cdot \int_0^y \int_x^{h+k_1-1} (y-x)^{\ell-k_1-1} e^{-\left(\frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{\beta'} - s\right)x} e^{-y\left(\frac{1}{\beta'} - t\right)} dx dy$$

and with the aid of Erdelyi (1954, p 187) (3.2) becomes

$$(3.3) \quad E^I[e^{sX+tY}] = \frac{\alpha^{-h}(\beta')^{-\ell}}{\Gamma(h)} \sum_{k_1=0}^{\ell-1} \frac{(\frac{\beta'}{\beta})^{k_1}}{k_1!} \frac{\Gamma(h+k_1)}{\Gamma(h+\ell)} \\ \cdot \int_0^y y^{h+\ell-1} e^{-y\left(\frac{1}{\beta'} - t\right)} {}_1F_1[h+k_1; h+\ell; -\left(\frac{1}{\alpha} + \frac{1}{\beta} - \frac{1}{\beta'} - s\right)y] dy$$

Expand ${}_1F_1(\cdot)$ as a power series in (3.3) to solve the integral. In a similar way $E^{II}[e^{sX+tY}]$ follows when $0 < y < x$. Thus when $(\frac{1}{\beta'} - t) > 0$ and $(\frac{1}{\alpha} - s) > 0$

$$M_{X,Y}(s,t) = \frac{\alpha^{-h}}{\Gamma(h)} \sum_{k_1=0}^{\ell-1} \frac{\Gamma(h+k_1)}{k_1!} \left(\frac{1}{\beta'}\right)^{k_1} (1-\beta't)^{-\ell+k_1} \left(\frac{1}{\alpha} + \frac{1}{\beta} - s - t\right)^{-(h+k_1)} \\ + \frac{\beta^{-\ell}}{\Gamma(\ell)} \sum_{k_2=0}^{h-1} \frac{\Gamma(\ell+k_2)}{k_2!} \left(\frac{1}{\alpha}\right)^{k_2} (1-\alpha's)^{-h+k_2} \left(\frac{1}{\alpha} + \frac{1}{\beta} - s - t\right)^{-(\ell+k_2)}$$

The general moment about the origin is:

$$\mu'_{r_1 r_2} = \frac{\partial^{r_1+r_2}}{\partial s^{r_1} \partial t^{r_2}} \left[M_{X,Y}(s,t) \right]_{s=0, t=0}$$

In particular the first and second order moments about the origin are

$$\mu'_{10} = \frac{\beta^{-\ell}}{\Gamma(\ell)} \sum_{k_2=0}^{h-1} \frac{\Gamma(\ell+k_2)}{k_2!} \left(\frac{1}{\alpha}\right)^{k_2} \alpha'(h-k_2) \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^{-(\ell+k_2)}$$

$$\begin{aligned}
& + \frac{\beta^{-\ell}}{\Gamma(\ell)} \frac{h-1}{k_2 \sum_{k_2=0}^{\ell-1}} \frac{\Gamma(\ell+k_2+1)}{k_2!} \left(\frac{1}{\alpha}\right)^{k_2} \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^{-(\ell+k_2+1)} \\
& + \frac{\alpha^{-h}}{\Gamma(h)} \frac{\ell-1}{k_1 \sum_{k_1=0}^{\ell-1}} \frac{\Gamma(h+k_1+1)}{k_1!} \left(\frac{1}{\beta}\right)^{k_1} \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^{-(h+k_1+1)}
\end{aligned}$$

and by symmetry μ'_{01} follows from μ'_{10} .

$$\begin{aligned}
\mu'_{20} = & \frac{\alpha^{-h}}{\Gamma(h)} \frac{\ell-1}{k_1 \sum_{k_1=0}^{\ell-1}} \frac{\Gamma(h+k_1+2)}{k_1!} \left(\frac{1}{\beta}\right)^{k_1} \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^{-(h+k_1+2)} \\
& + \frac{\beta^{-\ell}}{\Gamma(\ell)} \frac{h-1}{k_2 \sum_{k_2=0}^{\ell-1}} \frac{\Gamma(\ell+k_2+2)}{k_2!} \left(\frac{1}{\alpha}\right)^{k_2} \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^{-(\ell+k_2+2)} \\
& + 2 \frac{\beta^{-\ell}}{\Gamma(\ell)} \frac{h-1}{k_2 \sum_{k_2=0}^{\ell-1}} \frac{\Gamma(\ell+k_2+1)}{k_2!} \left(\frac{1}{\alpha}\right)^{k_2} (h-k_2) \alpha' \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^{-(\ell+k_2+1)} \\
& + \frac{\beta^{-\ell}}{\Gamma(\ell)} \frac{h-1}{k_2 \sum_{k_2=0}^{\ell-1}} \frac{\Gamma(\ell+k_2)}{k_2!} \left(\frac{1}{\alpha}\right)^{k_2} (h-k_2)(h-k_2+1) (\alpha')^2 \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^{-(\ell+k_2)}
\end{aligned}$$

and by symmetry μ'_{02} follows from μ'_{20} .

$$\begin{aligned}
\mu'_{11} = & \frac{\alpha^{-h}}{\Gamma(h)} \frac{\ell-1}{k_1 \sum_{k_1=0}^{\ell-1}} \frac{\Gamma(h+k_1+2)}{k_1!} \left(\frac{1}{\beta}\right)^{k_1} \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^{-(h+k_1+2)} \\
& + \frac{\alpha^{-h}}{\Gamma(h)} \frac{\ell-1}{k_1 \sum_{k_1=0}^{\ell-1}} \frac{\Gamma(h+k_1+1)}{k_1!} \left(\frac{1}{\beta}\right)^{k_1} (\ell-k_1) \beta' \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^{-(h+k_1+1)} \\
& + \frac{\beta^{-\ell}}{\Gamma(\ell)} \frac{h-1}{k_2 \sum_{k_2=0}^{\ell-1}} \frac{\Gamma(\ell+k_2+2)}{k_2!} \left(\frac{1}{\alpha}\right)^{k_2} \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^{-(\ell+k_2+1)} \\
& + \frac{\beta^{-\ell}}{\Gamma(\ell)} \frac{h-1}{k_2 \sum_{k_2=0}^{\ell-1}} \frac{\Gamma(\ell+k_2+1)}{k_2!} \left(\frac{1}{\alpha}\right)^{k_2} (h-k_2) \alpha' \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^{-(\ell+k_2+1)}
\end{aligned}$$

Remarks:

1. Finding the expressions for the moments of interest is tedious. However these ex-

pressions have the advantage that once they have been obtained they are easily computed for specific values of the parameters involved.

2. From the above moments the variances and the correlation coefficient, ρ , easily follow. Limits for ρ are obtained from two extreme cases which are just excluded from the model. The upper limit, which is equal to 1, follows when $\alpha' \rightarrow 0$ and $\beta' \rightarrow 0$ i.e. the system fails when one component fails. For $\alpha = \beta$ the lower limit,

$$\rho_0 = \frac{\sum_{k_2=0}^{h-1} \sum_{k_1=0}^{\ell-1} \frac{\Gamma(\ell+k_2)\Gamma(h+k_1)}{\Gamma(h)\Gamma(\ell)} \frac{(h-k_2)(\ell-k_1)}{k_1! k_2!} 2^{-(h+\ell+k_1+k_2)}}{\left\{ \sum_{k_2=0}^{h-1} \frac{\Gamma(\ell+k_2)(h-k_2)(h-k_2+1)}{k_2! \Gamma(\ell)} 2^{-(\ell+k_2)} - \left(\sum_{k_2=0}^{h-1} \frac{\Gamma(\ell+k_2)(h-k_2)}{k_2! \Gamma(\ell)} 2^{-(\ell+k_2)} \right)^2 \right\} \left\{ \sum_{k_1=0}^{\ell-1} \frac{\Gamma(h+k_1)(\ell-k_1)(\ell-k_1+1)}{k_1! \Gamma(h)} 2^{-(h+k_1)} - \left(\sum_{k_1=0}^{\ell-1} \frac{\Gamma(h+k_1)(\ell-k_1)}{k_1! \Gamma(h)} 2^{-(h+k_1)} \right)^2 \right\}}^{\frac{1}{2}}$$

follows when $\alpha' \rightarrow \infty$ and $\beta' \rightarrow \infty$ i.e. once one component has failed the other keeps on functioning forever.

4. Parameter Estimation

Consider a sample (x_i, y_i) $i=1, 2, \dots, N$ from a bivariate distribution with density function (2.7). Assume that in the first r ($0 < r < N$) cases C_1 fails before C_2 i.e. C_1 receives its h^{th} shock before C_2 receives its ℓ^{th} shock and in the remaining $N-r$ cases C_2 fails before C_1 .

$$\text{Let } \underline{\theta} = \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} \text{ where } \theta_1 = \begin{pmatrix} \alpha \\ \alpha' \end{pmatrix} \text{ and } \theta_2 = \begin{pmatrix} \beta \\ \beta' \end{pmatrix},$$

then from (2.7) the likelihood function may now be written as

$$(4.1) \quad L(\underline{\theta}) = \frac{\alpha^{-rh} (\alpha')^{-(N-r)}}{[\Gamma(h) \Gamma(\ell)]^r} \prod_{i=r+1}^N y_i^{\ell-1} \left[\frac{1}{\alpha'} (x_i - y_i) + \frac{1}{\alpha} y_i \right]^{h-1} \cdot e^{-\frac{1}{\alpha'} \sum_{i=r+1}^N (x_i - y_i)} e^{-\frac{1}{\alpha} \sum_{i=1}^N \min(x_i, y_i)}$$

$$\begin{aligned} & \cdot \frac{\beta^{-\ell(N-r)}(\beta')^{-r}}{[\Gamma(h) \Gamma(\ell)]^{N-r}} \prod_{i=1}^r x_i^{h-1} \left[\frac{1}{\beta'}(y_i - x_i) + \frac{1}{\beta} x_i \right]^{\ell-1} \\ & \cdot e^{-\frac{1}{\beta'} \sum_{i=1}^r (y_i - x_i)} e^{-\frac{1}{\beta} \sum_{i=1}^N \min(x_i, y_i)} \\ & = V_1(\underline{\theta}_1) \cdot V_2(\underline{\theta}_2) \end{aligned}$$

so that

$$\begin{aligned} \frac{\partial \ln V_1(\underline{\theta}_1)}{\partial \alpha} &= \frac{-rh}{\alpha} + (h-1) \sum_{i=r+1}^N \frac{-y_i/\alpha^2}{\left[\frac{1}{\alpha'}(x_i - y_i) + \frac{1}{\alpha} y_i \right]} + \frac{1}{\alpha^2} \sum_{i=1}^N \min(x_i, y_i); \\ \frac{\partial \ln V_1(\underline{\theta}_1)}{\partial \alpha'} &= \frac{-(N-r)}{\alpha'} + (h-1) \sum_{i=r+1}^N \frac{-(x_i - y_i)/(\alpha')^2}{\left[\frac{1}{\alpha'}(x_i - y_i) + \frac{1}{\alpha} y_i \right]} + \frac{1}{(\alpha')^2} \sum_{i=r+1}^N (x_i - y_i); \\ \frac{\partial \ln V_2(\underline{\theta}_2)}{\partial \beta} &= \frac{-\ell(N-r)}{\beta} + (\ell-1) \sum_{i=1}^r \frac{-x_i/\beta^2}{\left[\frac{1}{\beta'}(y_i - x_i) + \frac{1}{\beta} x_i \right]} + \frac{1}{\beta^2} \sum_{i=1}^N \min(x_i, y_i); \\ \frac{\partial \ln V_2(\underline{\theta}_2)}{\partial \beta'} &= \frac{-r}{\beta'} + (\ell-1) \sum_{i=1}^r \frac{-(y_i - x_i)/(\beta')^2}{\left[\frac{1}{\beta'}(y_i - x_i) + \frac{1}{\beta} x_i \right]} + \frac{1}{(\beta')^2} \sum_{i=1}^r (y_i - x_i). \end{aligned}$$

The likelihood equations obtained by equating these derivatives to zero do not have a solution for $\underline{\theta}$ in closed form. No element of $\underline{\theta}$ can be expressed in any other way but as a function of itself and the other element of $\underline{\theta}$, in which it appears. An iterative procedure will have to be employed to solve for $\hat{\underline{\theta}}$, i.e. that value of $\underline{\theta}$ which maximises (4.1), and in particular Newton-based methods (Lawley and Maxwell, 1971, appendix II) are considered, that is

$$(4.2) \quad \hat{\underline{\theta}}^n = \hat{\underline{\theta}}^p - \epsilon H^{\star-1} \underline{g}$$

and ϵ is usually chosen to maximise the rate of convergence. In (4.2) $\hat{\theta}^n$ represents a new approximation of $\hat{\theta}$ while $\hat{\theta}^p$ denotes the previous approximation.

$$\underline{g} = \frac{\partial \ln L}{\partial \underline{\theta}} \Big|_{\underline{\theta} = \hat{\theta}^p} \text{ and for the Newton-Raphson method } H^* \text{ is the hessian}$$

$$H = \frac{\partial^2 \ln L}{\partial \underline{\theta} \partial \underline{\theta}'} \Big|_{\underline{\theta} = \hat{\theta}^p} \text{ which must be negative definite.}$$

For determining H it is convenient to replace $\underline{\theta}'$ by $\hat{\theta}^{*'} = (\lambda_1, \lambda'_1, \lambda_2, \lambda'_2)$ where $\lambda_1 = \frac{1}{\alpha}$, $\lambda'_1 = \frac{1}{\alpha'}$, $\lambda_2 = \frac{1}{\beta}$, $\lambda'_2 = \frac{1}{\beta'}$ since from the reparameterisation theorem (Graybill, 1961, p. 36) $\hat{\theta}$ can be obtained from $\hat{\theta}^*$. The hessian is

$$H(\theta) = \frac{\partial^2 \ln L}{\partial \underline{\theta}^* \partial \underline{\theta}^{*'}} = \begin{pmatrix} H_1(\underline{\theta}_1^*) & 0 \\ 0 & H_2(\underline{\theta}_2^*) \end{pmatrix}$$

where

$$H_1(\underline{\theta}_1^*) = \begin{pmatrix} -\frac{rh}{\lambda_1^2} & -\sum_{i=r+1}^N \frac{(h-1)y_i^2}{[\lambda'_1(x_i - y_i) + \lambda_1 y_i]^2} & -\sum_{i=r+1}^N \frac{(h-1)y_i(x_i - y_i)}{[\lambda'_1(x_i - y_i) + \lambda_1 y_i]^2} \\ -\sum_{i=r+1}^N \frac{(h-1)y_i(x_i - y_i)}{[\lambda'_1(x_i - y_i) + \lambda_1 y_i]^2} & -\frac{N-r}{(\lambda'_1)^2} & -\sum_{i=r+1}^N \frac{(h-1)(x_i - y_i)^2}{[\lambda'_1(x_i - y_i) + \lambda_1 y_i]^2} \end{pmatrix}$$

$$H_2(\underline{\theta}_2^*) = \begin{pmatrix} -\frac{(N-r)\ell}{\lambda_2^2} & -\sum_{i=1}^r \frac{(\ell-1)x_i^2}{[\lambda'_2(y_i - x_i) + \lambda_2 x_i]^2} & -\sum_{i=1}^r \frac{(\ell-1)x_i(y_i - x_i)}{[\lambda'_2(y_i - x_i) + \lambda_2 x_i]^2} \\ -\sum_{i=1}^r \frac{(\ell-1)x_i(y_i - x_i)}{[\lambda'_2(y_i - x_i) + \lambda_2 x_i]^2} & -\frac{r}{(\lambda'_2)^2} & -\sum_{i=1}^r \frac{(\ell-1)(y_i - x_i)^2}{[\lambda'_2(y_i - x_i) + \lambda_2 x_i]^2} \end{pmatrix}$$

and by solving $|H_1(\underline{\theta}_1^*) - \xi I| = 0$ and $|H_2(\underline{\theta}_2^*) - \xi I| = 0$ it can be shown that the characteristic roots ξ_1 and ξ_2 of $H_1(\underline{\theta}_1^*)$ and ξ_3 and ξ_4 of $H_2(\underline{\theta}_2^*)$ respectively are always negative so that $H(\underline{\theta}^*)$ is negative definite for all $\underline{\theta}^*$.

Solving for $\underline{\theta}^*$ can be simplified considerably by exploiting the fact that $L(\underline{\theta}^*) = V_1(\underline{\theta}_1^*)V_2(\underline{\theta}_2^*)$, and $L(\underline{\theta}^*)$ is maximised by the separate maximisation of $V_1(\underline{\theta}_1^*)$ and $V_2(\underline{\theta}_2^*)$. This maximisation is done by replacing $\underline{\theta}$ with $\underline{\theta}_1^*$ and H^* by $-H_1(\underline{\theta}_1^*)$ in the Newton-Raphson method.

The Fletcher-Powell iterative procedure described by Lawley and Maxwell (1971) involves much simpler computations than the Newton-Raphson method and may also be employed to find $\underline{\theta}^*$. By making use of the information matrix Fletcher-Powell's first step will be the same as that of Newton-Raphson, and in every further iteration the positive definite matrix required is obtained by updating that of the previous iteration while the Newton-Raphson method has to calculate this matrix anew for each iteration.

It has been established that the matrix
$$\left[\frac{\partial^2 \ln L(\underline{\theta}^*)}{\partial \underline{\theta}^* \partial \underline{\theta}^{*'}} \right]$$
 is negative definite

for all $\underline{\theta}^*$ so that (Kendall and Stuart, 1961, section 18.22) the vector of maximum likelihood estimators is unique and consequently (Wilks 1962, theorem 12.7.3) has an asymptotic quadrivariate normal distribution with covariance matrix the inverse of the information matrix.

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Manuscript received 1979-08-07, revised 1980-10-08.